

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<reasoning process here>

To find the de Broglie wavelength of an electron with a kinetic energy of1 eV , we use the de Broglie wavelength formula :

$\lambda = h / p$

where :

- h is Planck’s constant = 6.626×10^{-34} J·s
- p is the momentum of the electron

We are given kinetic energy $K = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

For non-relativistic electrons , the kinetic energy is related to momentum by :

$K = p^2 / (2m)$

where m is the electron mass = 9.109×10^{-31} kg

Solving for p :

$p = \sqrt{2mK}$

Then substitute into de Broglie formula :

$\lambda = h / \sqrt{2mK}$

Plug in values :

$\lambda = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$

Compute denominator :

$2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} = 2.918 \times 10^{-49}$

$\sqrt{2.918 \times 10^{-49}} \approx 5.402 \times 10^{-25}$

Now compute λ :

$\lambda = 6.626 \times 10^{-34} / 5.402 \times 10^{-25} \approx 1.227 \times 10^{-9} \text{ m}$

Now convert to cm : $1 \text{ m} = 100 \text{ cm}$ $\implies 1.227 \times 10^{-9} \text{ m} = 1.227 \times 10^{-7} \text{ cm}$

So $\lambda \approx 1.23 \times 10^{-7} \text{ cm}$

Now look at the options :

- A. 2.48×10^{-8} cm $\implies$ too small
- B. 5.92×10^7 cm/sec $\implies$ units are wrong
- C. 8.66×10^{-6} cm $\implies$ too large
- D. 3.14×10^{-5} cm $\implies$ too large
- E. 9.81×10^{-4} cm $\implies$ too large
- F. 5.27×10^{-9} m $\implies 5.27 \times 10^{-11}$ cm $\implies$ wrong value and units
- G. 1.23×10^{-7} cm $\implies$ matches our result
- H. 4.14×10^{-10} m $\implies 4.14 \times 10^{-8}$ cm $\implies$ not correct
- I. 6.625×10^{-27} erg·sec $\implies$ Planck’s constant in erg·s $\implies$ not a wavelength
- J. 0.512×10^6 eV $\implies$ energy , not wavelength

Option G : 1.23×10^{-7} cm $\implies$ this is equivalent to 1.23×10^{-9} m $\implies$ matches computed value .

Therefore , correct answer is G .

</reasoning process>

<answer>G</answer>